

MUN: an AI-powered multiplayer networking solution for VR games

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ABSTRACT

We present a new toolset named MUN (Metaverse Unity Networking) based on AI algorithms for creating multiplayer VR environments, developed on the basis of user-centered design research. It includes a Unity asset enabling creation and configuration of the multiplayer mode in a VR setup, supplemented with 1) a recommendation engine for optimal programming methods, with error detection in application source code, 2) a user behavior analytics system in multiplayer mode, enhancing realism in avatar and NPC actions using a motion atlas and motion recognition. Developed using Python 3.9, .NET, and Unity v. 2021, it was validated by Unity developers through three rounds of usability tests (N=64).

Index Terms: Virtual reality—Multiplayer video game—Artificial Intelligence—Unity; Asset—Motion recognition—Motion Atlas—User-centered design

1 INTRODUCTION

As the Metaverse evolves, there is an increasing need for user-friendly tools that support the development and optimization of multiplayer solutions [8, 9]. The main challenges and difficulties associated with implementing multiplayer modes in virtual reality (VR) games include ensuring seamless and accurate synchronization of players' movements and interactions, managing network latency and bandwidth constraints, addressing the physical and ergonomic considerations of VR hardware, and ensuring a consistent and stable gaming experience across different VR systems. Within Unity game engine, there are several multiplayer solutions available. Unity's own networking libraries, such as Netcode solutions [7, 9], as part of Unity Gaming Services, offer a broad spectrum of support for various game genres and provide cross-platform connectivity. For VR specifically, the Unity SharedSpaces Sample [6], provides a framework for creating simple multiplayer VR applications using Unity. Photon Unity Networking (PUN, [5]) emerges as a significant option that is highly flexible and supports cross-platform multiplayer experiences. Another notable solution is Mirror [1], which is popular in the Unity community for VR and non-VR projects alike. Each of these solutions has its own strengths and can be chosen based on the specific needs of the project, such as the scale of the multiplayer experience, the complexity of VR interactions, and the level

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Figure 1: A visualization of motion recognition engine - when the user's gesture is recognized, the category label is presented in interface, and the virtual mirror shows the detected movement.

of support required during development. Unity's in-house solutions offer a comprehensive and integrated approach, while third-party tools provide flexibility and community-driven support, which can be particularly useful for indie developers or those with more specialized requirements. The motivation of this work was to facilitate and accelerate the multiplayer development process using dedicated AI solutions. In the initial phase of the project, we conducted analysis with the use of Internet ethnography of over 500 posts from 6 sources commonly used by developers seeking technical assistance, including Unity forums, Discord servers, Reddit, etc. Content analysis was performed using a categorization key to tag and categorize the most frequently reported problems. Additionally, we performed qualitative studies with eight individual interviews and one focus group interview among Unity developers to define their needs and expectations about the new asset.

2 THE SOLUTION

The key feature of the MUN solution is the semi-automation of the multiplayer mode deployment process in VR games. This solution enables developers to implement multiplayer functionality more quickly and efficiently, reducing development time and costs. The MUN project also includes the development of a motion library, which is a crucial component in creating realistic VR experiences, and it can significantly reduce the data needed to be transmitted during multiplayer mode synchronization. As a result, developers

will have access to a rich collection of natural movements, significantly enhancing the quality and realism of interactions in the virtual world. The core functionalities of the MUN asset are divided into several modules. The Client Application allows developers to create their multiplayer solution, serving as a crucial asset. The Service for User and Server Management stores information about users, organizations, applications, and servers, facilitating communication with the Server Manager. The Server Manager manages servers on a specific device, while the MasterServer represents the target server for potential users, acting as the central server in the multiplayer asset framework. The asset involves several key functionalities, starting with joining the MasterServer. Users can create and join rooms, manage a list of available rooms, and supervise users within a room. It also includes the creation and management of network objects, as well as the synchronization of parameters, including character movement. The system supports Remote Procedure Calls (RPC) and features a voice chat option, allowing verbal communication between participants.

2.1 AI-based recommendation mode

An artificial intelligence engine developed to automate some functionalities of the asset is responsible for recognizing and suggesting the next steps for the programmer. It is also able to recognize and repair some of the developer's errors. These modules work in tandem to streamline the development process, enhancing efficiency and reducing the likelihood of mistakes in the programming workflow. The recommendation module was trained on a previously trained GPT-2 model using 2,7 million lines of code. It was tested on a data set comprising 891,880 lines of code and achieved an effectiveness of more than 70%. The ability to detect errors in application code and suggest corrective actions was assessed using a set of 1,005 general coding error examples with their corresponding repair methods, as well as 279 examples specifically related to errors in VR source code. The most effective algorithm used, LightGBM, achieved an accuracy of 91.05%.

2.2 Automatic movement detection and prediction mode based on motion atlas

Another AI engine was trained to recognize user movements. The system's input consists of the current position and rotation of key body points: the head, left hand, and right hand. A data buffer is used to analyze collected data in a temporal context. The motion recognition module, at its output, delivers the name of the identified movement, which is then relayed to the motion prediction module. This module has its own data buffer, which stores a history of recently performed movements. Based on this data buffer and the newly recognized movement, the motion prediction module returns a likely movement that the VR headset user will perform next. The system's output comprises information about the current movement being performed (Fig. 1) and the predicted next movement of the user. In order to implement the AI engine for motion recognition, a new and unique motion atlas has been developed, featuring 56 different movements specifically designed for VR games. To identify and recognize these motion patterns, 300 different models were developed using advanced machine learning techniques. Of these, the most efficient models were chosen for implementation. The atlas achieved a remarkable weighted F-score of nearly 98% in motion recognition, demonstrating its effectiveness [10]. The final database of movements of over 74 users will be made publicly available to the scientific community. To measure the prediction accuracy, a data set was prepared using recordings of sequences performed by 221 players, resulting in a total of 1,279 attempts to recognize the next movement. Each player performed a single sequence consisting of 6 movements. On this test set, the developed solution achieved a result of 88.51% accuracy.

3 DEMONSTRATION GAMES

For testing and demonstration purposes, two prototype multiplayer games with distinct features were developed using Unity version 2021.3.18f1. Players require VR goggles compatible with SteamVR to fully experience the games. Additionally, a functioning mandatory connection to SteamVR is essential, with active motion tracking in the headset and connected controllers to ensure seamless interaction within the virtual environments. The first game, the *MUN FPS Prototype*, is an interactive VR first-person shooter (FPS) developed using the Unity Engine. This project focuses on providing a dynamic and engaging multiplayer gameplay experience, where players can move, shoot, and compete in a virtual environment. The second game, the *MUN Survival Prototype* is a VR survival game in which players explore an island, gather resources, build, and fight for survival. This project integrates survival elements with the capabilities of VR interaction.

4 EVALUATION

Comprehensive UX and usability studies were conducted on the developed asset with the participation of 64 developers, in 3 iterations, using a protocol that encompassed both qualitative and quantitative methods, including eye tracking. Average System Usability Scale score was over 60 in the last iteration of the user research.

5 CONCLUSION

The project aims to automate and facilitate the creation of multiplayer VR games, supporting Metaverse development. The implementation of libraries/plugins that provide the tools developed in the form of a Unity asset is accessible through [2]. This setup ensures seamless integration and accessibility for users who wish to utilize these advanced functionalities in their Unity projects. The motion atlas can be accessed at [3]. Detailed descriptions of the project are available at [4]. Future directions of the project include further automation of the developer's work pipelines, enhancement of the motion recognition accuracy and scalability, i.e. by providing the possibility to add new custom movements to the atlas and retrain the recognition system.

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